

SHAYAN SHEKARFOROUSH

Google Scholar $\diamond$ Webpage

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EDUCATION

University of Toronto

2020 - Present

Direct PhD in Computer Science

GPA: 4/4

Supervisory Committee: David Fleet, Marcus Brubaker, David Lindell

Sharif University of Technology

2015 - 2020

Bachelor's of Science in Computer Engineering

GPA: 4/4 (19.58 / 20)

Minor in Mathematics

PUBLICATIONS

1. CryoSPIN: Improving Ab-Initio Cryo-EM Reconstruction with Semi-Amortized Pose Inference

Shayan Shekarforoush, David Lindell, Marcus Brubaker, David Fleet

NeurIPS 2024

2. Dual Camera Joint Deblurring-Denoising

Shayan Shekarforoush, Amanpreet Walia, Marcus Brubaker, Konstantinos Derpanis, Alex Levinstein

ArXiv

3. Residual Multiplicative Filter Networks for Multiscale Reconstruction

Shayan Shekarforoush, David Lindell, David Fleet, Marcus Brubaker

NeurIPS 2022

4. Physics Aware Inference for the Cryo-EM Inverse Problem

Geoffrey Woollard, Shayan Shekarforoush, Frank Wood, Marcus Brubaker, Khanh Dao Duc

NeurIPS Workshop 2022

5. Graph Convolution Based Attention Model for Personalized Disease Prediction

A. Kazi, S. Shekarforoush, S.A Krishna, H. Burwinkel, G. Vivar, B. Wiestler, K. Kortuem, SA. Ahmadi, S. Albarqouni, and N. Navab

MICCAI 2019

6. InceptionGCN: Receptive Field Aware Graph Convolutional Network for Disease Prediction

A. Kazi, S. Shekarforoush, S.A. Krishna, H. Burwinkel, G. Vivar, K. Kortuem, SA. Ahmadi, S. Albarqouni, and N. Navab

IPMI 2019

Oral Presentation (7% Acceptance rate)

7. Self-Attention Equipped Graph Convolutions for Disease Prediction

A. Kazi, S.A. Krishna, S. Shekarforoush, K. Kortuem, S. Albarqouni, and N. Navab

ISBI 2019

Oral Presentation

HONORS AND AWARDS

- **Vector Research Grant | \$6000 per year** 2021-2024
- **Mitacs Accelerate, Samsung AI and University of Toronto | \$21,000** Summer 2022
- **PIMS Travel Award** May 2022
Giving a talk the workshop of Mathematical and Computational Challenges in Cryo-EM in Banff
- **International Undergrad Excellence Award | €4,500** Summer 2018, 2019
Awarded by Technical University of Munich to strongly-motivated international undergraduate students interested in machine learning, medical imaging, and computer vision.
- **Ranked 1st based on GPA among Bachelors of Computer Engineering** Summer 2020
- **Ranked 92nd in National Universities Entrance Exam** Summer 2015

RESEARCH COLLABORATIONS

PhD Intern

July 2022 - April 2023

Samsung AI Center Toronto

Mentor: Alex Levinshtein

Visiting Student

July 2018 - July 2020

Computer Aided Medical Procedures & Augmented Reality Lab (CAMPAR)

Technical University of Munich (TUM), Supported by a scholarship from TUM

Supervisor: Nassir Navab.

Remote Collaboration

December 2018 - March 2019

European Molecular Biology Laboratory (EMBL)

Supervisor: Laleh Haghighverdi. (Erwin Schrodinger Prize winner)

Research Assistant

December 2018 - February 2020

Machine Learning Lab (MLL)

Sharif University of Technology, Iran

Supervisor: Marzieh Soleymani.

RELATED COURSES

Graduate

- **Geometry Processing (A+)**
- **Current Algorithms in 3D Vision (A+)**
- **Introduction to Machine Learning (A+)**
- **Probabilistic Learning and Reasoning (A+)**

Undergrad

- **Differential Geometry (19/20)**
- **Stochastic Processes (20/20)**
- **Information Theory (20/20)**
- **Linear Algebra (19.8/20)**

VOLUNTEER EXPERIENCE

Senior Scientific Member

December 2018 - March 2019

Sharif Datadays National Competition, Tehran, Iran
(more than 500 participating teams)

- Our team designed some tasks that could be solved by Machine Learning and Deep Learning techniques, based on advertising data gathered from Divar platform.
- This was the first Data Analytics event held by Sharif University of Technology consisting of different stages. Its main goal was to broaden the appeal to work on problems related to data analytics. It was also a large-scale attempt to assess the level of knowledge in this field in Iran.

TECHNICAL SKILLS

Programming Languages	Python, C/C++, Matlab, Java, Julia
Machine Learning Libraries	PyTorch, JaX, Tensorflow
Operating Systems	OS X, Linux, Windows
Miscellaneous	Blender, PostgreSQL, Git, Terminal, L ^A T _E X, Amazon AWS

LANGUAGES

Persian	Native
English	Advanced proficiency